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ANTIBIOTICS IN AGROECOSYSTEMS: STATE OF THE SCIENCE

Tetracycline and Phenicol Resistance Genes and Mechanisms: Importance for Agriculture, the Environment, and Humans

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Abstract

Recent reports have speculated on the future impact that antibiotic-resistant bacteria will have on food production, human health, and global economics. This review examines microbial resistance to tetracyclines and phenicols, antibiotics that are widely used in global food production. The mechanisms of resistance, mode of spread between agriculturally and human-impacted environments and ecosystems, distribution among bacteria, and the genes most likely to be associated with agricultural and environmental settings are included. Forty-six different tetracycline resistance (*tet*) genes have been identified in 126 genera, with *tet*(M) having the broadest taxonomic distribution among all bacteria and *tet*(B) having the broadest coverage among the Gram-negative genera. Phenicol resistance genes are organized into 37 groups and have been identified in 70 bacterial genera. The review provides the latest information on tetracycline and phenicol resistance genes, including their association with mobile genetic elements in bacteria of environmental, medical, and veterinary relevance. Knowing what specific antibiotic-resistance genes (ARGs) are found in specific bacterial species and/or genera is critical when using a selective suite of ARGs for detection or surveillance studies. As detection methods move to molecular techniques, our knowledge about which type of bacteria carry which resistance gene(s) will become more important to ensure that the whole spectrum of bacteria are included in future surveillance studies. This review provides information needed to integrate the biology, taxonomy, and ecology of tetracycline- and phenicol-resistant bacteria and their resistance genes so that informative surveillance strategies can be developed and the correct genes selected.

Core Ideas

- Updates resistance genes that are important in the environment, agriculture, and humans.
- Suggests best genes to use for screening the environment.
- Provides correct nomenclature for tetracycline and phenicol resistance genes.
- Provides tables that will be useful to researchers.

The discovery and use of antibiotics for the treatment of animal, plant, and human bacterial infections represents one of the most significant discoveries of the 20th century. Antibiotics now save millions of animal and human lives annually, reduce losses to agricultural and horticultural crops, and contribute to increased food productivity. Widespread use of antibiotics during the last 70 yr, however, has led to increases in the level of antibiotic-resistant bacteria (ARB) and antibiotic-resistance genes (ARGs) isolated in agriculture and aquaculture settings, the environment, and humans (Centers for Disease Control and Prevention, 2013; Van Boeckel et al., 2015; World Health Organization, 2014). These findings have led some researchers to anticipate a "post-antibiotic era," when animals, plants, and people die of common infections that were once easily treatable with antibiotics (Collignon, 2013).

Concerns about the spread of antibiotic resistance have fueled many groups to assess the impacts of ARBs and ARGs on human and animal health, agricultural and food production, and agriculture, aquaculture, and human waste management (Akinbowale et al., 2007; Centers for Disease Control and Prevention, 2013; The White House, 2014; Taylor et al., 2014; World Health Organization, 2014). One of the primary outcomes of these reports is a call for increased surveillance of ARBs and ARGs in agricultural and environmental settings, with a particular interest in identifying transmission routes of ARBs and ARGs throughout the world (Collignon, 2013; The White House, 2014). Key to the success of current and future surveillance efforts are strategies to identify which particular types of resistance genes to measure and the validity of monitoring the presence of gene sequences without phenotype assessment. Antibiotic-resistance genes are not randomly distributed among bacteria. There is a clear link between bacterial taxonomy and specific types of ARGs (Durso et al., 2012; Roberts et al., 2012) (Tables 1 and 2), a phenomenon that has been particularly well documented with tetracycline resistance (Chopra and Roberts, 2001; Roberts, 2012; Roberts et al., 2015; Sloan et al.,

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Abbreviations: ABC, adenosine triphosphate binding cassette; ARB, antibiotic-resistant bacteria; ARG, antibiotic-resistance gene; bp, base pair; CAT, chloramphenicol acetyltransferase; Cm^r, chloramphenicol-resistant; EF-Tu, elongation factor thermo unstable; G+C, guanine + cytosine; GTP, guanosine triphosphate; ICE, integrative and conjugative element; MFS, major facilitator superfamily; MIC, minimal inhibitory concentration; Tc^r, tetracycline-resistant; TMS, transmembrane segment.

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Table 1. Distribution of *tet* and *otr* resistance genes (adapted from Roberts et al., 2012; http://faculty.washington.edu/marilynr/).

Gene	Gram-positive bacteria, *Mycobacterium*, *Mycoplasma*, *Nocardia*, *Streptomyces*, and *Ureaplasma* ($n = 50$)	Gram-negative bacteria ($n = 76$)	Total genera ($n = 126$)	Genera
	no.			
				Mechanism: Efflux ($n = 30$)
tet(A)	0	23	23	*Acinetobacter, Aeromonas, Alcaligenes, Bordetella, Chryseobacterium, Citrobacter, Edwardsiella, Enterobacter, Escherichia, Flavobacterium. Klebsiella, Laribacter, Ochrobactrum, Plesiomonas, Proteus, Pseudomonas, Rhizobium, Salmonella, Serratia, Shigella Variovorax, Veillonella, Vibrio*
tet(B)	0	33	33	*Acinetobacter, Actinobacillus, Aeromonas, Aggregatibacter, Brevundimonas, Citrobacter, Enterobacter, Erwinia, Escherichia, Gallibacterium, Haemophilus, Klebsiella, Mannheimia, Moraxella, Neisseria, Ochrobactrum, Pantoea, Pasteurella, Plesiomonas, Photobacterium, Proteus, Providencia, Pseudoalteromonas, Pseudomonas, Roseobacter, Rhizobium, Salmonella, Serratia, Shewanella, Shigella, Treponema, Vibrio, Yersinia*
tet(C)	0	16	16	*Aeromonas, Bordetella, Chlamydia, Citrobacter, Enterobacter, Escherichia, Francisella, Halomonas, Klebsiella, Proteus, Pseudomonas, Roseobacter, Salmonella, Serratia, Shigella, Vibrio*
tet(D)	0	22	22	*Aeromonas, Alteromonas, Brevundimonas, Burkholderia, Chryseobacterium, Citrobacter, Edwardsiella, Enterobacter, Escherichia, Halomonas, Klebsiella, Morganella, Pasteurella, Plesiomonas, Photobacterium, Pseudomonas, Rhizobium, Salmonella, Shewanella, Shigella, Vibrio, Yersinia*
tet(E)	0	10	10	*Aeromonas, Alcaligenes, Escherichia, Flavobacterium, Proteus, Providencia, Pseudomonas, Roseobacter, Serratia, Vibrio*
tet(G)	0	15	15	*Acinetobacter, Brevundimonas, Enterobacter, Escherichia, Fusobacterium, Mannheimia, Ochrobactrum, Pasteurella, Proteus, Providencia, Pseudomonas, Roseobacter, Salmonella, Shewanella, Vibrio*
tet(H)	0	12	12	*Acinetobacter, Actinobacillus, Aeromonas, Bibersteinia, Gallibacterium, Gilliamella, Histophilus, Mannheimia, Moraxella, Pasteurella, Proteus, Psychrobacter*
tet(J)	0	3	3	*Escherichia, Morganella, Proteus*
tet(K)	13	3	16	*Bacillus, Clostridium, Enterococcus, Eubacterium, Gallibacterium, Haemophilus, Lactobacillus, Listeria, Mycobacterium, Nocardia, Peptostreptococcus, Sporosarcina, Staphylococcus, Streptococcus, Streptomyces, Vibrio*
tet(L)	22	19	41	*Acinetobacter, Actinobacillus, Actinomyces, Bacillus, Bifidobacterium, Bhargavaea, Citrobacter, Clostridium, Enterobacter, Enterococcus, Escherichia, Flavobacterium, Fusobacterium, Gallibacterium, Geobacillus, Kurthia, Lactobacillus, Listeria, Mannheimia, Morganella, Mycobacterium, Nocardia, Oceanobacillus, Ochrobactrum, Paenibacillus, Pasteurella, Pediococcus, Peptostreptococcus, Proteus, Pseudomonas, Rahnella, Salmonella, Savagea, Sporosarcina, Staphylococcus, Streptococcus, Streptomyces, Vagococcus, Variovorax, Veillonella, Virgibacillus*
tetA(P)	1	0	1	*Clostridium*
tet(V)	1	0	1	*Mycobacterium*
tet(Y)	0	3	3	*Aeromonas, Escherichia, Photobacterium*
tet(Z)	2	0	2	*Corynebacterium, Lactobacillus*
tet(30)	0	1	1	*Agrobacterium*
tet(31)	0	3	3	*Aeromonas, Bordetella, Gallibacterium*
tet(33)	2	0	2	*Arthrobacter, Corynebacterium*
tet(35)	0	2	2	*Stenotrophomonas, Vibrio*
tet(38)	1	0	1	*Staphylococcus*
tet(39)	4	8	12	*Acinetobacter, Alcaligenes, Bacillus, Brevundimonas, Cellulosimicrobium, Corynebacterium, Enterobacter, Lysinibacillus, Providencia, Pseudomonas, Psychrobacter, Stenotrophomonas*
tet(40)	2	0	2	*Clostridium, Streptococcus*
tet(41)	0	1	1	*Serratia*
tet(42)	5	1	6	*Bacillus, Microbacterium, Micrococcus, Paenibacillus, Pseudomonas, Staphylococcus*
tet(43)	1	0	1	*Staphylococcus*
tet(45)	2	0	2	*Bacillus, Bhargavaea*
*tet*AB(46)	1	0	1	*Streptococcus*

Table 1. Continued.

Gene	Gram-positive bacteria, *Mycobacterium*, *Mycoplasma*, *Nocardia*, *Streptomyces*, and *Ureaplasma* ($n = 50$)	Gram-negative bacteria ($n = 76$)	Total genera ($n = 126$)	Genera
	no.			
tet(47)	0	1	1	*Providencia*
tcr	1	0	1	*Streptomyces*
otr(B)	2	0	2	*Mycobacterium*, *Streptomyces*
otr(C)	1	0	1	*Streptomyces*
				Mechanism: Ribosomal protection ($n = 12$)
tet(M)	38	37	75	*Abiotrophia*, *Acinetobacter*, *Actinomyces*, *Aerococcus*, *Aeromonas*, *Afipia*, *Amycolatopsis*, *Anaerococcus*, *Arthrobacter*, *Bacteroides*, *Bacterionema*, *Bacillus*, *Bifidobacterium*, *Bhargavaea*, *Brachybacterium*, *Catenibacterium*, *Citrobacter*, *Clostridium*, *Corynebacterium*, *Edwardsiella*, *Eikenella*, *Enterobacter*, *Enterococcus*, *Escherichia*, *Eubacterium*, *Erysipelothrix*, *Finegoldia*, *Flavobacterium*, *Fusobacterium*, *Gardnerella*, *Gemella*, *Granulicatella*, *Hafnia*, *Haemophilus*, *Helcococcus*, *Kingella*, *Klebsiella*, *Kurthia*, *Lawsonia*, *Lactobacillus*, *Lactococcus*, *Leifsonia*, *Listeria*, *Microbacterium*, *Mycobacterium*, *Mycoplasma*, *Neisseria*, *Norcardia*, *Paenibacillus*, *Pantoea*, *Pasteurella*, *Peptostreptococcus*, *Photobacterium*, *Prevotella*, *Proteus*, *Providencia*, *Pseudoalteromonas*, *Pseudomonas*, *Psychrobacter*, *Ralstonia*, *Rahanella*, *Rhizobium*, *Savagea*, *Selenomonas*, *Serratia*, *Shewanella*, *Shigella*, *Stenotrophomonas*, *Sporosarcina*, *Staphylococcus*, *Streptococcus*, *Streptomyces*, *Ureaplasma*, *Veillonella*, *Vibrio*
tet(O)	18	17	35	*Actinobacillus*, *Aerococcus*, *Aeromonas*, *Anaerovibrio*, *Bifidobacterium*, *Brevundimonas*, *Burkholderia*, *Butyrivibrio*, *Campylobacter*, *Citrobacter*, *Clostridium*, *Enterococcus*, *Eubacterium*, *Fusobacterium*, *Gemella*, *Granulicatella*, *Lactobacillus*, *Leifsonia*, *Megasphaera*, *Microbacterium*, *Mobiluncus*, *Mycobacterium*, *Neisseria*, *Norcardia Paenibacillus*, *Pasteurella*, *Peptostreptococcus*, *Pseudomonas*, *Psychrobacter*, *Rhizobium*, *Rhodococcus*, *Shewanella*, *Staphylococcus*, *Stenotrophomonas*, *Streptococcus*
tetB(P)	1	0	1	*Clostridium*
tet(Q)	8	11	19	*Anaerovibrio*, *Bacteroides*, *Capnocytophaga*, *Clostridium*, *Eubacterium*, *Fusobacterium*, *Gardnerella*, *Lactobacillus*, *Mitsuokella*, *Mobiluncus*, *Neisseria*, *Peptostreptococcus*, *Porphyromonas*, *Prevotella*, *Ruminococcus*, *Selenomonas*, *Streptococcus*, *Subdoligranulum*, *Veillonella*
tet(S)	6	3	9	*Citrobacter*, *Enterococcus*, *Klebsiella*, *Lactobacillus*, *Lactococcus*, *Listeria*, *Staphylococcus*, *Streptococcus*, *Veillonella*
tet(T)	2	7	9	*Aeromonas*, *Brevundimonas*, *Chryseobacterium*, *Enterococcus*, *Pseudomonas*, *Rhizobium*, *Shewanella*, *Stenotrophomonas*, *Streptococcus*
tet(W)	10	22	32	*Acidaminococcus*, *Acinetobacter*, *Actinomyces*, *Arcanobacterium*, *Bacillus*, *Bacteroides*, *Bifidobacterium*, *Brevundimonas*, *Butyrivibrio*, *Chryseobacterium*, *Citrobacter*, *Clostridium*, *Corynebacterium*, *Escherichia*, *Fusobacterium*, *Klebsiella*, *Lactobacillus*, *Lawsonia*, *Megasphaera*, *Mitsuokella*, *Neisseria*, *Porphyromonas*, *Prevotella*, *Pseudomonas*, *Rhizobium*, *Roseburia*, *Selenomonas*, *Shewanella*, *Staphylococcus*, *Streptococcus*, *Subdolgranulum*, *Veillonella*
tet(32)	2	0	2	*Eubacterium*, *Streptococcus*
tet(36)	2	1	3	*Bacteroides*, *Clostridium*, *Lactobacillus*
tet(44)	2	1	3	*Campylobacter*, *Clostridium*, *Staphylococcus*
tet	1	0	1	*Streptomyces*
otr(A)	3	0	3	*Bacillus*, *Mycobacterium*, *Streptomyces*
				Carrying mosaic genes
	4	1	5	*Bifidobacerium tet*(O/W/32/O/W/O), *tet*(W/32/O), *tet*(O/W) *Clostridium tet*(O/32/O) *Streptococcus tet*(O/W/32/O), *tet*(O/32/O), *tet*(S/M) *Lactobacillus tet*(W/32/O/W/O), *Megasphaera tet*(O/W), *tet*(O/W/O)
				Mechanism: Enzymatic ($n = 3$)
tet(X)	0	11	11	*Comamonas*, *Bacteroides*, *Delftia*, *Enterobacter*, *Escherichia*, *Klebsiella*, *Riemerella*, *Pseudomonas*, *Spingobacterium*, *Serratia*, *Wautersiella*
tet(34)	0	4	4	*Aeromonas*, *Pseudomonas*, *Serratia*, *Vibrio*
tet(37)	0	0	0	Uncultured
				Mechanism unknown ($n = 1$)
tet(U)	3	0	3	*Enterococcus*, *Staphylococcus*, *Streptococcus*

Table 2. Acquired phenicol resistance genes (last update: 19 Apr. 2015; adapted from Schwarz et al. [2004]**,** Shen et al. [2013]**, and nucleotide BLAST searches).**

Group	Gene	Gene(s) included	Gram-positive bacteria, *Streptomyces*	Gram-negative bacteria	Total genera	Genera
			no.			
				Mechanism: Efflux (major facilitator superfamily)		
Type E-1	*cmlA1*	*cmlA*, *cmlA2*, *cmlA4*, *cmlA5*, *cmlA6*, *cmlA7*, *cmlA8*, *cmlA10*, *cmlB*	2	15	17	*Acinetobacter*, *Aeromonas*, *Arcanobacterium*, *Citrobacter*, *Comamonas*, *Enterobacter*, *Escherichia*, *Klebsiella*, *Laribacter*, *Pantoea*, *Providencia*, *Pseudomonas*, *Ralstonia*, *Salmonella*, *Serratia*, *Shigella*, *Staphylococcus*
Type E-2	*cml*			1	1	*Escherichia*
Type E-3	*floR*	*cmlA*-like, *flo*, *pp-flo*, *cmlA9*, *floRv*		22	22	*Acinetobacter*, *Actinobacillus*, *Aeromonas*, *Azospirillum*, *Bibersteinia*, *Bordetella*, *Edwardsiella*, *Enterobacter*, *Escherichia*, *Kaistia*, *Klebsiella*, *Mannheimia*, *Pasteurella*, *Photobacterium*, *Providencia*, *Pseudomonas*, *Riemerella*, *Rhodospirillum*, *Salmonella*, *Shinella*, *Stenotrophomonas*, *Vibrio*
Type E-4	*fexA*	*fexAv*	6		6	*Bacillus*, *Enterococcus*, *Exiguobacterium*, *Nosocomiicoccus*, *Staphylococcus*, *Streptococcus*
Type E-5	*cml*		1		1	*Streptomyces*
Type E-6	*cmlv*		1		1	*Streptomyces*
Type E-7	*cmrA*	*cmr*	1		1	*Rhodococcus*
Type E-8	*cmr*	*cmx*	1		1	*Corynebacterium*
Type E-9	*cmlB1*			1	1	*Bordetella*
Type E-10	*fexB*		1		1	*Enterococcus*
Type E-11	*pexA*					uncultured
				Mechanism: Efflux (ATP-binding cassette)		
Type D-1	*optrA*		1		1	*Enterococcus*
				Mechanism: Inactivating enzyme		
Type A-1	*catA1*	*cat*, *catI*, *pp-cat*	4	18	22	*Achromobacter*, *Acinetobacter*, *Aeromonas*, *Bifidobacterium*, *Citrobacter*, *Clostridium*, *Corynebacterium*, *Enterobacter*, *Escherichia*, *Flavobacterium*, *Herbaspirillum*, *Klebsiella*, *Morganella*, *Neisseria*, *Parabacteroides*, *Photobacterium*, *Proteus*, *Ruminococcus*, *Salmonella*, *Serratia*, *Shigella*, *Yersinia*
Type A-2	*catA2*	*cat*, *catII*		15	15	*Aeromonas*, *Agrobacterium*, *Avibacterium*, *Citrobacter*, *Enterobacter*, *Escherichia*, *Gallibacterium*, *Haemophilus*, *Legionella*, *Klebsiella*, *Morganella*, *Photobacterium*, *Proteus*, *Salmonella*, *Vibrio*
Type A-3	*catA3*	*cat*, *catIII*		10	10	*Actinobacillus*, *Bibersteinia*, *Edwardsiella*, *Escherichia*, *Haemophilus*, *Klebsiella*, *Mannheimia*, *Pasteurella*, *Providencia*, *Shigella*
Type A-4	*cat*			1	1	*Proteus*
Type A-5	*cat*		1		1	*Streptomyces*
Type A-6	*cat86*		1		1	*Bacillus*
Type A-7	cat_{pC221}	*cat*, *catC*	6	1	7	*Bacillus*, *Enterococcus*, *Gallibacterium*, *Lactobacillus*, *Listeria*, *Staphylococcus*, *Streptococcus*
Type A-8	cat_{pC223}	*cat*	6		6	*Bacillus*, *Enterococcus*, *Lactococcus*, *Listeria*, *Staphylococcus*, *Streptococcus*
Type A-9	cat_{pC194}	*cat*, *cat-TC*	7	1	8	*Bacillus*, *Enterococcus*, *Gallibacterium*, *Lactobacillus*, *Paenibacillus*, *Ruminiclostridium*, *Staphylococcus*, *Streptococcus*
Type A-10	*cat*		1		1	*Bacillus*
Type A-11	*catP*	*catD*	1	1	2	*Clostridium*, *Neisseria*
Type A-12	*catS*		1	1	2	*Haemophilus*, *Streptococcus*
Type A-13	*cat*			1	1	*Campylobacter*
Type A-14	*cat*			5	5	*Aeromonas*, *Listonella*, *Photobacterium*, *Proteus*, *Vibrio*
Type A-15	*catB*		1		1	*Clostridium*
Type A-16	*catQ*		1		1	*Clostridium*
Type B-1	*catB1*	*cat*		1	1	*Agrobacterium*
Type B-2	*catB2*			11	11	*Acinetobacter*, *Aeromonas*, *Bordetella*, *Citrobacter*, *Enterobacter*, *Escherichia*, *Klebsiella*, *Pasteurella*, *Proteus*, *Pseudomonas*, *Salmonella*
Type B-3	*catB3*	*catB4*, *catB5*, *catB6*, *catB8*	1	22	23	*Achromobacter*, *Acinetobacter*, *Aeromonas*, *Bordetella*, *Citrobacter*, *Comamonas*, *Delftia*, *Enterobacter*, *Escherichia*, *Klebsiella*, *Kluyvera*, *Morganella*, *Pantoea*, *Proteus*, *Providencia*, *Pseudomonas*, *Raoultella*, *Riemerella*, *Salmonella*, *Serratia*, *Staphylococcus*, *Stenotrophomonas*, *Vogesella*
Type B-4	*catB7*			1	1	*Pseudomonas*
Type B-5	*catB9*			1	1	*Vibrio*
Type B-6	*catB10*			1	1	*Pseudomonas*

Table 2. Continued.

Group	Gene	Gene(s) included	Gram-positive bacteria, *Streptomyces*	Gram-negative bacteria	Total genera	Genera
			no.			
						Mechanism: rRNA methylase
	cfr		6	2	8	*Bacillus*, *Enterococcus*, *Escherichia*, *Jeotgalicoccus*, *Macrococcus*, *Proteus*, *Staphylococcus*, *Streptococcus*
	cfr(B)	*cfr*-like, *cfr*	2		2	*Clostridium*, *Enterococcus*
						Mechanism: Hydrolase
	estDL136					uncultured

1994). In the context of the transfer of resistance from food animals to humans, it has been suggested that the same gene might have different risks of environmental transmission depending on the specific taxa carrying the gene (Durso et al., 2012). Knowing that specific ARGs are found in only some bacterial species and/or genera is critical when using a selective suite of ARGs for regional, national, and international surveillance studies. This review looks at the historical and current use of tetracyclines and phenicols and focuses on measuring ARB and ARGs associated with the use of these antibiotics in food production systems around the globe.

The rationale for including both tetracyclines and phenicols in this review is because of the differences in application during the past 40 yr in livestock, humans, small animals, agriculture, and aquaculture. Tetracyclines have been an important antibiotic group and have continued to be used in livestock, agriculture, aquaculture, and small animal and human medicine since its introduction (Food and Drug Administration, 2014; Roberts, 2003, 2005a). Tetracyclines have also been used as food additives (41% in US food in 2012) in the United States since the 1950s, although with new Food and Drug Administration (FDA) guidelines, this may change the levels of tetracyclines used in the future (Federal Register, 2013; Food and Drug Administration, 2014). In contrast, the use of chloramphenicol has been limited in humans in the United States, the European Union (EU), and other industrial countries, although it continues to be used in the developing world for the treatment of severe diarrhea and pneumonia because it is so inexpensive and stable. Thus, in the United States, EU, and industrialized countries, chloramphenicol is primarily used in the treatment of pets and non-food-producing animals, which may be an important reservoir for resistant bacteria (Schwarz et al., 2004). Florfenicol is exclusively licensed for use in food-producing animals and may allow those chloramphenicol resistance genes that also confer cross-resistance to florfenicol to flourish and spread to other bacterial ecosystems (Schwarz et al., 2004).

Tetracycline Resistance

One of the early antibiotic classes discovered was tetracyclines, which were isolated from different soil *Streptomycetes* spp. between 1947 and the 1950s, with newer derivatives isolated between 1966 and 1972 (Levy, 2002, p. 15–56). Tetracyclines are broad-spectrum antibiotics that have activity against a wide range of Gram-positive and Gram-negative bacteria, atypical bacteria such as chlamydiae, mycoplasmas, and rickettsiae, and some protozoan parasites such as *Plasmodium* spp. causing malaria. Extensive use of tetracycline in production of food animals increased dramatically between the 1950s and 1970s, and by 2001, eightfold more antibiotics were used for food animals in the United States than for human use (11.2 million kg in animal agriculture and 1.4 million kg in human medicine) (Mellon et al., 2001). Today, tetracyclines are still the second most commonly used antibiotics in US agriculture (Food and Drug Administration, 2014). However, bacterial resistance to these antibiotics has reduced their effectiveness, as illustrated by the situation in Japan. In 1950, chloramphenicol, streptomycin, and tetracycline were introduced to Japan for treatment of *Shigella* infections (Falkow, 1975, p. 1–7). By 1953, both streptomycin-resistant (Sm^r) and tetracycline-resistant (Tc^r) *Shigella* spp. were observed, while the first chloramphenicol-resistant (Cm^r) *Shigella* was isolated in 1956. By 1960, 10.6% of the *Shigella* spp. were Tc^r, with 90% of them being multi-drug resistant (Cm^r, Tc^r, and Sm^r) (Falkow, 1975, p. 1–7). These Tc^r strains represented a number of different serotypes, indicating this was not the spread of a single clone. Chloramphenicol-resistant, Tc^r, and Sm^r *Escherichia coli* were also identified in patients with Cm^r, Tc^r, and Sm^r *Shigella* spp. infections. Within 8 yr of the discovery of resistant isolates in Japan, similar antibiotic-resistant *Salmonella enterica* subsp. *enterica* serovar Typhimurium and other Gram-negative bacteria were identified around the world (Falkow, 1975, p. 1–7).

At the same time, tetracyclines were used not only for treating animal infections but also as "growth promoters" or "feed efficiency products" added at subtherapeutic levels to feed to improve how quickly animals put on weight with the same amount of food (Chopra and Roberts, 2001). These practices were stopped in the EU by 1 Jan. 2006, although they were banned earlier in Scandinavian countries (European Commission, 2005). The practice continues in the United States today. For example, in 2011, 12.2 million kg of antimicrobials were sold for use in animals in the United States, of which 42.2% were tetracyclines (Food and Drug Administration, 2014). Tetracycline sold for use in animals may end up as environmental contaminants via excretion of unmetabolized antibiotics (Agwuh and MacGowan, 2006). In the United States, practices may be changing because in Nov 2013 the Food and Drug Administration (Federal Register, 2013) put forth a plan to phase out the use of growth-promoting antibiotics in livestock feed. The phase-in of this change is scheduled to occur over a 3-yr time period. The FDA plan calls on animal pharmaceutical companies to voluntarily change the labels on antibiotic feed products to eliminate their FDA approval as growth promoters, although relabeling the antibiotics as "preventive drugs" is not prohibited (Federal Register, 2013). Food and Drug Administration rules would also change the status from over-the-counter to prescription for antibiotics, which would mean that before obtaining antibiotics for treatment, a farmer would need to get a veterinarian to

provide a prescription. Whether the FDA ruling will reduce the use of subtherapeutic levels in healthy food animals is not clear at this time. Perhaps more significantly, the restaurant and supermarket industries in the United States are beginning to demand antibiotic-free products, which should impact the overall use of antibiotics in agriculture in the coming years.

Tetracyclines bind to the elongating ribosome and inhibit delivery of the ternary complex elongation factor thermo unstable (EF-Tu), guanosine triphosphate (GTP), and aminoacylated-tRNA to the A site (Chopra and Roberts, 2001; Donhofer et al., 2012; Roberts, 2005b). The primary binding site of tetracycline is located in the helix 34 of the 16S rRNA in the 30S subunit, which overlaps the anticodon stem loop of the A-site tRNA (Chopra and Roberts, 2001; Donhofer et al., 2012; Roberts, 2005b). During the last 70 yr, there has been widespread use of tetracycline in both animals and humans, which has led to an increase in tetracycline resistance. Tetracycline resistance is most often due to the acquisition of new genes that code for energy-dependent efflux of tetracyclines ($n = 30$ different genes), proteins that protect bacterial ribosomes from the action of tetracyclines ($n = 12$ different genes), or enzymatically inactivate tetracyclines ($n = 3$ different genes); for one gene, the mechanism of action is still unknown (Table 3). Many of these genes are associated with mobile plasmids, transposons, and conjugative transposons. These elements often code for their own transfer, which may influence their distribution among various genera and partially explain the wide differences in host range among the various genes (Table 1). Integrons, which are gene-capture systems and allow multiple ARGs to be linked, may be found in many of the genera listed in Table 1, but *tet* genes are not generally associated with these elements (Chopra and Roberts, 2001; Roberts, 2005a).

A total of 46 different genes have been identified and characterized as conferring tetracycline resistance, and most are identified as *tet* genes (Table 3). In general, these genes confer resistance to chlortetracycline, doxycycline, oxytetracycline, and tetracycline, while resistance to the second-generation minocycline varies with the mechanism of resistance and there is limited resistance conferred to the newer third-generation glycylcyclines (Sader et al., 2013). The innate genes *otr*(A), *otr*(B), and *otr*(C) originally identified in *Streptomyces* spp. have a different nomenclature because they were the genes first identified in oxytetracycline-producing organisms and originally thought to be restricted to *Streptomyces* spp. However, now both *Bacillus* spp. and *Mycobacteria* spp. have been identified with *otr* genes (Table 1) (Roberts et al., 2012). In recent years, the identification of ARGs has expanded from the culture of humans and animals to also include the multiple different environments both prior and after the use of antibiotics (Bhullar et al., 2012; Perron et al., 2015), which has led to identification of the *tet* genes *tet*(38) to *tet*(47). These *tet* genes have been identified in environmental and animal- and aquaculture-associated bacteria (Roberts et al., 2015). This, along with the antibiotic resistome data, suggests that there are a large number of ARGs in the environment that have yet to be adequately described but clearly can be potential reservoirs for animal and human pathogens (Perry et al., 2014).

The ARGs can be distinguished from each other using molecular methods including DNA–DNA hybridization with oligonucleotide probes and DNA sequencing, microarrays, or more recently after the entire genome has been sequenced. Two genes are considered related, i.e., of the same class, and given the same gene designation if the gene products share $\geq$80% of the amino acid sequences in common with each other. Two genes are considered different from each other if their gene products share $\leq$79% amino acid sequence identity (Levy et al., 1999). This comparison can now be done using GenBank sequence information. However, during the last 10 yr, mosaic tetracycline resistance genes have been identified. These are interclass hybrid genes created by combining parts of two or three different ribosomal protection genes into a single functionally active gene (Levy et al., 2005; Stanton et al., 2005) (Table 1). The genes show different patterns of mosaicism but their final size remains the same (Palmieri et al., 2012). No other types of interclass hybrid *tet* genes have yet been identified in natural isolates.

Table 3. Mechanisms of resistance for characterized *tet* and *otr* genes.

Efflux ($n = 30$)	Ribosomal protection ($n = 12$)	Enzymatic ($n = 3$)	Unknown
tet(A), *tet*(B), *tet*(C), *tet*(D), *tet*(E)	*tet*(M), *tet*(O), *tet*(S), *tet*(W), *tet*(32)	*tet*(X)†	*tet*(U)‡
tet(G), *tet*(H), *tet*(J), *tet*(V), *tet*(Y)	*tet*(Q), *tet*(T), *tet*(36)	*tet*(34)	
tet(Z), *tet*(30), *tet*(31), *tet*(33), *tet*(47)	*otr*(A), *tetB*(P)§, *tet*	*tet*(37)†	
tet(35)¶	*tet*(44)		
tet(39), *tet*(41)			
tet (K), *tet*(L), *tet*(38), *tet*(45)			
tetA(P), *tet*(40)			
otr(B), *otr*(C)			
*tcr*3			
tet(42)			
tet(43)			
*tet*AB(46)#			

† *tet*(X) and *tet*(37) are unrelated but both are NADP-requiring oxidoreductases: *tet*(34) similar to the xanthine-guanine phosphoribosyl transferase genes of *Vibrio cholera*.

‡ *tet*(U) has been sequenced but does not appear to be related to either efflux or ribosomal protection proteins (http://faculty.washington.edu/marilynr/).

§ *tetB*(P) is not found alone and *tetA*(P) and *tetB*(P) are counted as one operon.

¶ Not related to other *tet* efflux genes.

Two genes needed for resistance *tet*AB(46) (http://faculty.washington.edu/marilynr/).

To identify a mosaic *tet* gene, the complete gene needs to be sequenced because a probe will not necessarily be able to distinguish a mosaic from a normal ribosomal protection gene (Levy et al., 2005; Palmieri et al., 2012; Stanton et al., 2005).

A total of 126 different genera including 76 Gram-negative and 50 Gram-positive have been identified as carrying known acquired tetracycline resistance genes (Table 1). The majority of new bacteria identified carrying *tet* genes have been from various environmental ecosystems (Table 1). A limited number of bacteria acquire tetracycline resistance by mutations, which alter the permeability of the outer membrane porins and/or lipopolysaccharides in the outer membrane, change regulation of innate efflux systems, or alter the 16S rRNA (Chopra and Roberts, 2001).

Chloramphenicol Resistance

Chloramphenicol was isolated from *Streptomyces venezuelae* in 1947 (Ehrlich et al., 1947) and shown to be a broad-spectrum antibiotic. Chloramphenicol was the first naturally occurring substance described to contain a nitro group. Its relatively simple structure made it the first antibiotic to be marketed as the product of chemical synthesis. Since 1950, chloramphenicol has been produced exclusively synthetically (Schwarz et al., 2004). Chloramphenicol is a highly specific and potent inhibitor of bacterial protein biosynthesis. Its bacteriostatic activity is based on a reversible binding to the peptidyl transferase center at the 50S ribosomal subunit of 70S ribosomes, where it prevents peptide chain elongation (Schlünzen et al., 2001). The 80S ribosomes of eukaryotic cells are not targets of chloramphenicol and its derivatives. However, it has been assumed that chloramphenicol may interact with mitochondrial ribosomes, which are similar in their structure to prokaryotic 70S ribosomes. As a potential consequence, the mitochondrial function of stem cells in bone marrow may be impaired, resulting in a suppression of the bone marrow function (Franklin and Snow, 1998). The hydroxyl group at C-3 in the chloramphenicol molecule can be replaced with fluorine to create the fluorinated thiamphenicol derivative florfenicol (Syriopoulou et al., 1981). In addition, the substitution of the nitro group ($-NO_2$), which is considered to be responsible for dose-unrelated aplastic anemia, by a sulfomethyl group ($-SO_2CH_3$) at the para position of the 1-phenyl moiety became effective in thiamphenicol and florfenicol (Schwarz et al., 2004).

Over the years, chloramphenicol and its derivatives, thiamphenicol and azidamfenicol, have been used extensively in human medicine. In the early years after its introduction into clinical use, chloramphenicol was considered a promising broad-spectrum antibiotic (Shaw, 1983). However, a number of adverse effects have been observed since the mid-1960s in connection with the use of chloramphenicol (Martelo et al., 1964): (i) a dose-unrelated irreversible aplastic anemia, which occurs at frequencies of 1:10,000 to 1:40,000 (Simon and Stille, 2000, p. 172–175) or 1:20,000 to 1:600,000 (Vinning and Stuttard, 1995), (ii) a dose-related reversible bone-marrow suppression, or (iii) the Gray syndrome in neonates and infants (Shaw, 1983; Yao and Moellering, 1999). Occasionally, hypersensitivity to chloramphenicol ranging from skin rashes to anaphylaxis has also been observed (Yao and Moellering, 1999). Due to these adverse effects and the availability of less toxic antimicrobial agents with a similar spectrum of activity, chloramphenicol therapy for humans is now limited to a small number of life-threatening infections (Yao and Moellering, 1999; Schwarz et al., 2004). For example, since chloramphenicol readily crosses the blood–brain barrier, it remains an alternative therapeutic agent for the treatment of meningitis caused by susceptible strains of *Haemophilus influenzae*, *Neisseria meningitis*, and *Streptococcus pneumoniae* when no other antimicrobial agents can be used, e.g., in penicillin-allergic patients (Mascaretti, 2003; Schwarz et al., 2004).

The use of chloramphenicol in veterinary medicine in the EU is currently limited to pets and non-food-producing animals. Chloramphenicol was banned in 1994 from use in any food-producing animals in the EU. The main reason for this ban was protection of the consumer from potential adverse effects arising from chloramphenicol residues in carcasses of food-producing animals. Because of the dose-independent irreversible aplastic anemia in humans, the "non-observed effect level" (NOEL) could not be determined for chloramphenicol (Schwarz et al., 2004). In toxicological studies, NOEL represents the dose at and below which adverse effects do not occur (Schwarz and Chaslus-Dancla, 2001). The calculation of the "maximum residue level," which represents the maximum level of antibiotic residues acceptable in carcasses at slaughter without any adverse effect on public health, is based on the NOEL and therefore could also not be determined. As a consequence, EU legislation banned chloramphenicol, along with several antibiotics, from use in food-producing animals (Schwarz et al., 2004). Because adverse side effects, in particular the dose-independent irreversible aplastic anemia, have not been observed in animals, the fluorinated thiamphenicol derivative florfenicol has been licensed, initially for the control of bacterial respiratory tract infections in cattle and pigs in 1995 and 2000, respectively. Moreover, florfenicol is also approved for the treatment of infectious pododermatitis (interdigital phlegmon) in cattle due to *Fusobacterium necrophorum* and *Bacteroides melaninogenicus* in the United States. In commercial salmon farming, a florfenicol premix is used for the treatment of furunculosis caused by *Aeromonas salmonicida*. Fluorinated phenicols are currently not used in human medicine (Schwarz et al., 2004).

The substrate spectrum of chloramphenicol includes Gram-positive and Gram-negative, aerobic and anaerobic bacteria, as well as chlamydiae, mycoplasmas, and rickettsiae (Shaw, 1983; Yao and Moellering, 1999). Chloramphenicol analogs including florfenicol have a similar spectrum of antimicrobial activity as chloramphenicol (Yao and Moellering, 1999). Intrinsic resistance to chloramphenicol and florfenicol has not been observed, although members of different bacterial species and genera may differ in their basic levels of susceptibility to both drugs, as confirmed by their different minimal inhibitory concentrations (MICs) (Yao and Moellering, 1999; Schwarz et al., 2004).

Chloramphenicol resistance is most often due to the presence of chloramphenicol acetyltransferases (CATs), which inactivate chloramphenicol (Murray and Shaw, 1997). There are two different types of CAT enzymes, which are genetically unrelated (Table 2). Chloramphenicol resistance may also be due to the efflux of chloramphenicol via specific membrane-associated transporters (Butaye et al., 2003). Both the genes coding for CATs and specific exporters are often associated with plasmids, transposons, or gene cassettes but also integrative and conjugative elements

(ICEs) or integrative and mobilizable elements (Schwarz et al., 2004). Some chromosomal multidrug transporters have also been identified that export chloramphenicol (Paulsen et al., 1996). Chloramphenicol resistance may also occur from mutations that reduce the expression of outer membrane proteins (Toro et al., 1990), mutations in the 23S rRNA (Ettayebi et al., 1985), inactivation of chloramphenicol by 3-*O*-phosphotransferases (Izard and Ellis, 2000), or target site modification by a 23S rRNA methylase (Kehrenberg et al., 2005; Long et al., 2006).

In contrast to the tetracycline-resistance genes, there is no internationally accepted nomenclature for chloramphenicol-resistance genes currently available. However, when using the same criteria as for the classification of *tet* genes, 16 groups of classical chloramphenicol acetyltransferase genes (*cat*), at least six groups of *cat* genes of the second type (which are occasionally referred to as *xat* genes [Murray and Shaw, 1997]), and 11 different groups of genes coding for specific exporters of the major facilitator superfamily (MFS) as well as a single ATP-binding cassette (ABC) transporter gene can be distinguished (Table 2). In addition, *cfr* and more recently *cfr*(B) have been characterized (Deshpande et al., 2015; Hansen and Vester, 2015; Marín et al., 2015). These genes code for the aforementioned rRNA methylases and mediate phenicol resistance by target site modification (Kehrenberg et al., 2005: Long et al., 2006; Hansen and Vester, 2015). Moreover, a single gene of unknown origin that confers chloramphenicol resistance by hydrolysis has been described (Tao et al., 2012). During whole genome sequencing, *cat*-like genes have been annotated in the genomes of several bacteria, e.g., *Brucella melitensis* (GenBank NC_003317) and *Bacillus cereus* (GenBank NC_004722). However, comparisons on both the nucleotide and amino acid levels revealed little, if any, homology with the known chloramphenicol-resistance determinants, and it has not been demonstrated whether these *cat*-like genes actually confer Cm^r. More information on phenicol resistance can be found in two reviews by Schwarz et al. (2004) and Schwarz and White (2005).

Mechanisms of Tetracycline Resistance

Tetracycline Resistance Due to Efflux Proteins

Most of the *tet*, *tcr*, and *otr* genes code for tetracycline resistance efflux proteins and are part of the MFS of transporters. The proteins are membrane bound and exchange a proton for a tetracycline–cation complex against a concentration gradient (Yamaguchi et al., 1990). This reduces the intracellular tetracycline concentration, which in turn protects the ribosomes within the cell. These proteins share amino acid and protein structure similarities with other efflux proteins involved in multiple-drug resistance, quaternary ammonium resistance, and chloramphenicol and quinolone resistance (Paulsen et al., 1996). The *tet*(B) gene found in Gram-negative bacteria codes for an efflux protein that confers resistance to chlortetracycline, doxycycline, minocycline, oxytetracycline, and tetracycline but generally not to tigecycline (Chopra and Roberts, 2001). All the other tetracycline efflux proteins confer resistance to tetracycline, chlortetracycline, doxycycline, and oxytetracycline but not to minocycline or tigecycline.

The majority of the *tet* efflux genes codes for an approximately 46 kDa membrane-bound efflux protein. With the exception of the *tet*AB(46), the efflux genes of Gram-negative bacteria in general have two functional domains, α and β, which correspond to the N- and C-terminal halves of the protein, respectively. Experimental data suggest that residues dispersed across the protein are important for function (McNicholas et al., 1995). Mutations affecting energy coupling have been located in cytoplasmic loops 2 to 3 and 10 to 11 of the efflux protein (McNicholas et al., 1992).

Many of the *tet* genes found in Gram-negative species that code for efflux have repressor genes upstream of the structural gene (Yamaguchi et al., 1990; McNicholas et al., 1995). In addition, the *tet*(Z) and *tet*(33) genes are of Gram-positive origin and code for efflux proteins that also have a repressor gene upstream of the structural gene. These were the first efflux proteins identified in Gram-positive bacteria that were shown to be controlled by a repressor protein (Tauch et al., 2000, 2002). The repressor genes are read in the opposite direction from the structural genes and regulate expression of the structural gene in the presence vs. absence of tetracycline. The majority of the 46 tetracycline-resistance efflux genes have been linked to other ARGs, such as those that confer aminoglycoside, β-lactam, chloramphenicol and florfenicol, macrolide, methicillin, streptomycin, sulfamethoxazole, or trimethoprim resistance or to genes coding for mercury resistance (Brenciani et al., 2004; Giovanetti et al., 2003; Roberts, 2012; Tauch et al., 2002). Selected *tet* genes are part of multiresistance elements, such as the integrative and mobilizable *Salmonella* genomic island 1 or ICE*Pmu1* from *Pasteurella multocida* and ICE*Mh1* from *Mannheimia haemolytica* (Roberts, 2012; Michael et al., 2012; Eidam et al., 2015).

In contrast, the *tet*(K) and *tet*(L) efflux genes of Gram-positive origin (guanine + cytosine [G+C] percentage of 28 to 40%), are not regulated by repressors but by translational attenuators, share 58 to 59% amino acid identity with each other, and confer resistance to tetracyclines but not minocycline. The *tet*(K) and *tet*(L) genes are generally found on small transmissible plasmids in Gram-positive bacteria. Occasionally, these plasmids or the *tet* genes have been associated with the chromosome. The *tet*(L) gene has been linked to the trimethoprim resistance gene *dfrK* (Kadlec and Schwarz, 2009; Roberts, 2012). A number of plasmid-borne *tet*(L) genes have been sequenced and share between 98 and 99% sequence identity with each other, while the chromosomal *tet*(L) gene from *Bacillis subtilis* has only 81% amino acid sequence identity with the other *tet*(L) genes (Stasinopoulos et al., 1998). Both *tet* genes were originally isolated in Gram-positive genera, but now the *tet*(K) gene has been isolated in four Gram-negative genera, while 19 out of 41 (48%) of the genera identified in Table 1 carrying the *tet*(L) gene are Gram-negative. The *tet*(K) gene has been linked to mercury resistance genes (Roberts, 2012).

The *tet*(39) gene was first isolated in Gram-negative bacteria and has a G+C percentage of 40%. It has now been isolated from Gram-positive genera, bringing into question the ancestral source of this gene (Adelowo and Fagade, 2009; Roberts et al., 2015). The *tet*(40) gene has a 42% G+C content and has been linked with the mosaic *tet*(O/32/O) gene (Roberts, 2012). The *tet*(43) gene also has a high G+C percentage of 65%, although it has only been identified in one Gram-positive genus (Table 1). The *tet*(45) has a G+C percentage of 34% and was found in Gram-positive bacteria (Table 1).

The TetP operon from *Clostridium perfringens* consists of two overlapping genes, *tetA*(P) (30% G+C) and *tetB*(P) (31%). The *tetA*(P) gene codes for an efflux protein but does not have the conserved motifs that are common in the other *tet* efflux proteins (Johanesen et al., 2001). The *tetA*(P) gene has been identified alone in isolates. In this case, there are some sequence differences between these *tetA*(P) gene sequences and those of the *tetA*(P) gene found with the overlapping *tetB*(P) gene (Sloan et al., 1994). The *tetB*(P) gene codes for a protein with amino acid identity of 37 to 39% with the Tet(M) ribosomal protection protein. The *tetA*(P) gene has been isolated from *Clostridium* spp. without the *tetB*(P) gene, but the reverse has not been found. The recently identified *tet*AB(46) gene codes for two non-identical proteins, and it has been suggested that they function as a heterodimeric ABC transporter requiring both proteins for function and tetracycline resistance and have 45% G+C content. There have not been any studies to verify physical interaction between the two proteins, but in-frame deletions of either *tet*A(46) or *tet*B(46) demonstrated that both proteins are needed for tetracycline resistance in the original host (Warburton et al., 2013). The newest *tet* gene is the *tet*(47) gene, which has been identified in the chromosome of a *Providencia rettgeri* isolate (GenBank WP_004258627).

Tetracycline Resistance Due to Ribosomal Protection Proteins

Twelve ribosomal protection proteins have been described (Table 2). These genes code for cytoplasmic proteins that protect the ribosomes from the action of tetracyclines except the glycylcycline tigecycline (Doan et al., 2006). Most studies have used the Tet(M) and/or Tet(O) proteins, which share ~75% sequence similarity and ~45% identity with elongation factor G (EF-G) (Chopra and Roberts, 2001; Donhofer et al., 2012; Connell et al., 2003). The tetracycline ribosomal protection proteins are grouped together within the translation factor superfamily of GTPases. The Tet(M) and Tet(O) proteins catalyze the release of tetracycline from the ribosomes in a GTP-dependent fashion. A similar mechanism is thought to occur with all the tetracycline ribosome protection proteins. Recently, using a cryo-EM structure of the Tet(M)-GDPNP-70S complex, it has been suggest that the Tet(M) protein confers resistance to tetracycline by directly interacting and altering the conformation of the nucleotide C1054 within helix 34 of the 16S rRNA, which is part of the tetracycline binding site. The altered conformation of the C1054 changes the stacking interaction with tetracycline, leading to dissociation from the ribosome, and prevents rebinding and promotes binding of the ternary complex EF-Tu–GTP–aa-tRNA (Donhofer et al., 2012). This mechanism works in vivo and in vitro, while the tetracycline-resistance efflux proteins require intact membranes to function. These *tet* genes may be evolutionarily derived from the elongation factors, but they cannot substitute for the elongation factors in vivo or in vitro (Connell et al., 2003).

The Tet(M), Tet(O), Tet(S), Tet(T), Tet(W), Tet(32), Tet(36), Tet(44) proteins share between 64 and 78% amino acid identity (Abril et al., 2010; Melville et al., 2001). The G+C content of seven of these genes varies from 30% [*tet*(44)] to 40%, and they are thought to be of Gram-positive origin. The exception is the *tet*(W) gene, which has a 53% G+C content (Roberts, 2005b), and the ancestral source of the *tet*(W) gene is unclear. All eight of these *tet* genes are found in both Gram-positive and Gram-negative genera (Table 1). The *tetB*(P) gene is less related than the other eight ribosomal protection *tet* genes. When the *tetB*(P) gene was cloned, alone it conferred on the host very low levels of tetracycline resistance (3 μg/mL) (Sloan et al., 1994). The genes, isolated originally from *Streptomyces* [*tet*, *otr*(A)], have higher G+C contents of >70%, as would be expected.

To date, all mosaic genes continue to be associated with rearrangements of ribosomal protection genes. Combinations of *tet*(O), *tet*(W), and/or *tet*(32) or *tet*(M) and *tet*(S) have been identified in Gram-positive bacteria, while combinations of *tet*(O) and *tet*(W) have been found in Gram-negative bacteria (Palmieri et al., 2012; Stanton and Humphrey, 2003) (Table 1). Why the mosaic ribosomal protection proteins are found in certain environments is not known, nor the advantages, if any, of carrying mosaic *tet* genes provides to its host.

The *tet*(M) gene is usually associated with conjugative transposons like the Tn*916*-Tn*1545* family (Clewell et al., 1995). This family of transposons includes small elements that just carry the *tet*(M) gene to large elements carrying multiple different ARGs including one or more of the following: the *erm*(B) gene, coding for an rRNA methylase, which confers macrolide, lincosamide, and streptogramin B resistance; *mef*(A); *msr*(D) coding for macrolide resistance, aminoglycoside phosphotransferase gene *aphA-3* encoding kanamycin resistance; *cat* genes coding for chloramphenicol resistance; *mer* operon coding for mercury resistance; the *erm*(B) gene; and other transposons such as Tn*917* (Roberts, 2005a, 2005b, 2012). The combination of *tet*(M) and *erm*(B) genes is often found in Gram-positive bacteria, such as streptococci, staphylococci, and enterococci (Roberts, 2005a, 2005b, 2012). The presence of the classical *cat* and *aphA-3* genes within these common transposons may explain why Cm^r and/or kanamycin-resistant (Km^r) *Streptococcus pneumoniae* strains are isolated in areas and from populations where the use of these antibiotics has been stopped (Luna and Roberts, 1998).

The *tet*(O) gene can be linked to the *mef*(A) and *msr*(D) macrolide-resistance genes. In *Streptococcus pyogenes* isolates from Italy, the *tet*(O) gene has been associated with a conjugative transposon that also carries a *mef*(A) gene that pumps erythromycin out of the cell (Brenciani et al., 2004; Giovanetti et al., 2003). The *tet*(Q) gene has been associated with the *erm*(F) gene, where it has been identified in a number of genera (Chung et al., 1999). The *tet*(Q) gene has also been linked to *erm*(B), *erm*(G), *mef*(A), and *msr*(D) macrolide-resistance genes in addition to *Bacteroides* conjugative transposons (Roberts, 2012).

Tetracycline Resistance Due to Enzymatic Inactivation

The *tet*(X) gene encodes an enzyme that modifies and inactivates the tetracycline molecule. It was originally found in a strict anaerobe, *Bacteroides*, where O_2 is excluded (Speer et al., 1991), linked to the *erm*(F) gene. It is unlikely that the *tet*(X) gene functions in *Bacteroides*, while the *erm*(F) gene does. The *tet*(X) gene has a G+C content of 38% and has now been found in 10 other Gram-negative genera (Table 1) where the gene can be expressed. The Tet(X) is a cytoplasmic protein that chemically modifies tetracycline in the presence of both O_2 and nicotinamide adenine dinucleotide phosphate (NADPH). Sequence analysis has

indicated that this protein shares amino acid homology with other NADPH-requiring oxidoreductases. A second gene, *tet*(37), has been identified from the oral human microbiome in two studies (Diaz-Torres et al., 2003, 2006). This protein also requires O_2 to function and should function in aerobic but not anaerobic species. Unfortunately, no attempt to determine the host(s) of this gene has been done (Table 1). The *tet*(34) gene was first identified in the chromosome of *Vibrio* spp. (Nonaka and Suzuki, 2002). The protein has homology to xanthine–guanine phosphoribosyl transferase genes, which act in purine nucleotide salvage synthesis. Nonaka and Suzuki (2002) suggested that the mechanism of the Tet(34) protein is the activation of Mg^{2+}–dependent purine nucleotide synthesis, which protects protein synthesis.

Other Unknown Mechanisms of Tetracycline Resistance

The *tet*(U) gene confers low-level tetracycline resistance (Chopra and Roberts, 2001). This gene produces a small protein (105 amino acids) that is smaller than the efflux and the ribosomal proteins. There is 21% similarity over the 105 amino acids between the Tet(U) and Tet(M) proteins beginning close to the carboxy terminus of the latter. These similarities do not include the consensus GTP-binding sequences, important for resistance in the ribosomal protection proteins. Recently, the *tet*(U) gene has been found in functional genomic assays, suggesting that it can confer tetracycline resistance under these types of laboratory conditions using 20 μg/mL tetracycline selection (Roberts et al., unpublished data, 2014). It has now been identified in three Gram-positive genera (Table 1). However, Caryl et al. (2012) suggested that the gene is a misannotated 3′ end of a gene encoding a rolling circle replication initiator (Rep) protein. Clearly more work needs to be done with this gene to determine if it confers tetracycline resistance and what the mechanism of that resistance is.

Tetracycline Resistance Due to Mutations

There is disagreement on whether direct mutations within the *tet*(A) and the *tet*(B) efflux genes are responsible for increased glycylcycline resistance or if it is due to mutations in other host genes (Fyfe et al., 2013; Akiyama et al., 2013). However, it is likely that with time and use, increased resistance may develop in species currently susceptible to tigecycline (Chopra, 2001). Tetracyclines show poor activity against some of the *Bacteroides fragilis* group of bacteria, and only 6% of the *Pseudomonas aeruginosa* were susceptible (Doan et al., 2006). Mutations that alter the permeability of the outer membrane porins and/or lipopolysaccharides in the outer membrane can also affect the bacterial host's resistance to tetracycline (Zhang et al., 2008). The Tc^r *Helicobacter pylori* and *Mycobacterium avium* complex and the anaerobic spirochete *Brachyspira hyodysenteriae* also have been identified with chromosomal mutations that increase tetracycline minimal inhibitory concentrations (Pringle et al., 2007; Toledo and Lopez-Solis, 2010).

Mutations that up-regulate innate efflux pumps can alter the host's susceptibility to tetracycline. One example is *Neisseria gonorrhoeae*, which has an innate *mtrCDE*-encoded efflux pump. A 1 base pair (bp) deletion of an A within the 13-bp inverted repeat sequence of the *mtrR* promoter region leads to a fourfold increased resistance to tetracycline, penicillin, and erythromycin (Cousin et al., 2003). In *N. gonorrhoeae*, chromosomally mediated resistance is often more common than plasmid-mediated antibiotic resistance (Centers for Disease Control and Prevention, 2002). Efflux pumps and their association with increased resistance to antibiotics including tetracycline have recently been reviewed by Sun et al. (2014).

Mechanisms of Chloramphenicol Resistance

Chloramphenicol Resistance Due to Chloramphenicol *O*-acetyltransferases

Both types of CAT enzymes have a trimeric structure, normally composed of three identical monomers, which is encoded by the *cat* gene (Murray and Shaw, 1997). The classical CAT monomers vary between 207 and 238 amino acids, whereas those of the second type of CATs are smaller, with 209 to 219 amino acids. All CATs transfer an acetyl group from a donor molecule (usually acetyl-CoA) to the C3 position of the chloramphenicol. This acetyl group is then shifted from C3 to C1, and the C3 position is again available for a second acetylation step. Neither the mono- nor the di-acetylated chloramphenicol molecules have antimicrobial activity (Murray and Shaw, 1997). None of the CAT enzymes are able to inactivate florfenicol, a thiamphenicol derivative that is exclusively licensed for use in animals (Schwarz and Chaslus-Dancla, 2001), because the C3 position is fluorinated in the florfenicol molecule. As a result, the C3 position of florfenicol cannot act as acceptor site for acetyl groups, making florfenicol resistant to inactivation by these enzymes.

The classical CATs represent a highly diverse group of enzymes that show an overall identity of 44%. These enzymes have been detected in Gram-positive and Gram-negative, aerobic and anaerobic bacteria (Table 2). They can be placed into 22 genetic groups, using ≥80% amino acid identity to define a group. However, 15 of the groups have a single gene from a single species (Table 2). The CATI, CATII, and CATIII groups, which represent members of the first three genetic groups in Table 2, are exclusively found in Gram-negative genera and are expressed constitutively. The genes coding for these enzymes have been completely sequenced and the biochemical and enzymatic characteristics of the proteins studied in detail (Shaw, 1983). The CATIII enzyme was the first to be crystallized and provided insight into the folding of the CAT monomers and helped to identify the amino acids that are important for the structure and the function of the CAT enzyme (Leslie et al., 1988).

The next three classical genetic CAT groups were named according to the plasmids (pC221, pC223/pSCS7, and pC194) on which they were first detected. These have been identified in a variety of Gram-positive genera (Table 2). The Michaelis constant (K_M) values for chloramphenicol and acetyl-CoA, the isoelectric point, pH optimum, and thermostability for the CATs associated with the Gram-positive plasmids have been determined (Cardoso and Schwarz, 1992; Shaw, 1983). These *cat* genes are induced by chloramphenicol and have translational attenuators located immediately upstream of the respective *cat* genes that resemble those located upstream of the tetracycline resistance genes *tet*(K) and *tet*(L) (Lovett, 1990).

The closely related CATP and CATD proteins were first identified in the Gram-positive anaerobe *Clostridium* spp., where they are located on transposons (Lyras and Rood, 2000). These genes have also been identified in Cm^r Gram-negative *Neisseria meningitidis* (Galimand et al., 1998; Shultz et al., 2003). This group is unusual because members are found in both Gram-positive and Gram-negative genera (Table 2). Both genes are expressed constitutively.

The second type of CAT enzymes are only distantly related to the classical CATs and are structurally similar to acetyltransferases involved in streptogramin A resistance (Murray and Shaw, 1997). At least five different genetic groups can be distinguished, although all the enzymes have approximately 77% identity with each other. These *cat* genes are often associated with gene cassettes and integrons in Gram-negative bacteria (Recchia and Hall, 1995). Some of these *cat* genes have also been identified in transposons. The CAT protein from *Agrobacterium tumefaciens* has different acetylation kinetics than the classical CATIII enzyme (Murray and Shaw, 1997). This difference might explain the distinctly lower chloramphenicol MIC mediated by this CAT protein. Because of this lower resistance level, it has been speculated that members of this second type of CATs might have a physiological role other than chloramphenicol resistance in their host bacteria, although little else is known about these enzymes (Murray and Shaw, 1997).

Phenicol Resistance Due to Specific Exporters

The specific exporters involved in the export of either chloramphenicol or chloramphenicol and florfenicol are mainly members of the MFS of efflux proteins (Paulsen et al., 1996) and commonly exhibit 10 to 14 transmembrane segments (TMSs) (Butaye et al., 2003). There are 11 genetic groups, although nine are found in a single genus including six from soil and environmental bacteria (Table 2). The *cmr* and *cmx* genes are found on plasmids, while the *cmx* gene is associated with a transposon (Butaye et al., 2003). The *Rhodococcus* genes are associated with plasmids, while the *cmrA* is located on transposon Tn*5561*. The *Streptomyces venezuelae* Cmlv protein is thought to play a role in self-defense of the antibiotic producer from its own products. Several closely related *cmlA* genes have been identified on gene cassettes in Gram-negative bacteria and, unlike other cassette-borne genes, *cmlA* is inducibly expressed by a translational attenuator, similar to that of staphylococcal *cat* genes (Recchia and Hall, 1995; Stokes and Hall, 1991). Another type of chloramphenicol exporter, CmlB1, which shares 74 to 77% identity with the known CmlA proteins, was identified on a plasmid from *Bordetella bronchiseptica* (Bissonnette et al., 1991; Kadlec et al., 2007). The *cmlB1* gene is also preceded by a translational attenuator and inducibly expressed. Neither the CmlA and CmlB1 proteins can efficiently export florfenicol from the bacterial cell, and bacteria with these genes are florfenicol susceptible.

Resistance to both chloramphenicol and florfenicol is characteristic for the group comprising *floR* and *pp-flo* genes. Butaye et al. (2003) reviewed these genes. The *pp-flo* gene was detected in the fish pathogen *Photobacterium damselae* subsp. *piscicida* (Kim and Aoki, 1996). The *floR* gene can be found in the chromosome of multi-resistant *Salmonella enterica* serovars including Typhimurium DT104, *Vibrio cholerae*, *E. coli*, *Bordetella bronchiseptica*, and *Acinetobacter baumannii*, or on plasmids of *E. coli*, *Salmonella* Newport, *Klebsiella pneumoniae*, *Pasteurella multocida*, *Bibersteinia trehalosi*, *Actinobacillus pleuropneumoniae*, and *Stenothrophomonas maltophila* (Schwarz et al., 2004). The *floR* gene has been identified as part of the chromosomally located ICE*Pmu1* from *Pasteurella multocida*, which carries and transfers a total of 12 antimicrobial resistance genes (Michael et al., 2012). A new *floR* gene variant (*floRv*), whose gene product showed only 84.1 to 91.8% amino acid identity to the various FloR proteins so far known, has recently been described to be part of a multi-drug-resistance genomic island in a *S. maltophilia* isolate of porcine origin (He et al., 2015).

The gene *fexA* is part of transposon Tn*558* and was first identified on a *Staphylococcus lentus* plasmid. Expression of *fexA* is inducible with either chloramphenicol or florfenicol. A translational attenuator similar to those of *cat* genes from *Staphylococcus* spp. and *Bacillus pumilus* was identified immediately upstream of the *fexA* gene (Kehrenberg and Schwarz, 2004). A *fexA* gene variant (*fexAv*), which conferred only chloramphenicol resistance, was recently identified in *Staphylococcus pseudintermedius* (Gómez-Sanz et al., 2013). The exporter FexAv exhibited two amino acid substitutions, Gly33Ala and Ala37Val, both of which seem to be important for substrate recognition. Site-directed mutagenesis that reverted the mutated base pairs to those present in the original *fexA* gene restored the chloramphenicol–florfenicol resistance phenotype. A novel florfenicol and chloramphenicol resistance gene, designated *pexA*, was discovered in Alaskan soil using a metagenomic approach (Lang et al., 2010). The gene *fexB*, which also confers resistance to chloramphenicol and florfenicol, was found on non-conjugative plasmids of *Enterococcus faecium* and *Enterococcus hirae* (Liu et al., 2012). It shows 59.7 and 37.2% nucleotide identity with the genes *fexA* and *pexA*, respectively, while the deduced FexB protein sequence exhibited 56.1 and 15.6% amino acid identity with the FexA and PexA protein sequences, respectively.

Florfenicol has been licensed in China since 1999 for the control of bacterial infections in cattle and chickens, and since then it has been used extensively in veterinary medicine and food animal production (Liu et al., 2012). Most recently, a novel gene, designated *optrA*, has been identified on a conjugative plasmid in *Enterococcus faecalis* (Wang et al., 2015). This gene codes for an ABC transporter that is able to confer resistance not only to chloramphenicol and florfenicol but also to the oxazolidinones linezolid and tedizolid. The *optrA* gene was functionally active in *E. faecalis*, *E. faecium*, and *Staphylococcus aureus*. A first survey conducted in China revealed that the *optrA* gene was more frequently found in enterococci of animal than of human origin (Wang et al., 2015).

Phenicol Resistance Due to Multidrug Transporters, Permeability Barriers, Mutations, Phosphorylation, Hydrolysis, or Target Site Methylation

Multidrug transporter systems assigned to the resistance–nodulation–cell division family have been reported to export phenicols from the bacterial cell and include the AcrAB-TolC system in *E. coli* (McMurry et al., 1994) and the MexAB-OprM and MexCD-OprJ systems in *Pseudomonas aeruginosa* (Paulsen et al., 1996). In Gram-positive bacteria, several 12-TMS multidrug transporters of the MFS, such as Blt and Bmr proteins

from *Bacillus subtilis* and NorA from *Staphylococcus aureus*, have been reported to have a substrate spectrum that includes chloramphenicol (Paulsen et al., 1996). Another two closely related 12-TMS multidrug efflux proteins, MdfA and Cmr, have been identified in *E. coli* (Edgar and Bibi, 1997; Nilsen et al., 1996).

The Cm^r Gram-negative bacteria may be due to the loss of or a distinct decrease in the expression of outer membrane proteins, which serve as the entry for chloramphenicol into the bacterial cell. Examples have been reported in *Haemophilus influenzae* (Burns et al., 1985), and *Salmonella enterica* serovar Typhi (Toro et al., 1990). Activation of the *mar* locus in *Enterobacteriaceae* may also play a role in the decreased expression of the outer membrane protein OmpF by producing an antisense RNA that interferes with the translation of *ompF* transcripts.

Several mutations in the 23S rRNA of *E. coli* (Ettayebi et al., 1985) are known to confer Cm^r. Deletions of 6 bp in the gene coding for the ribosomal protein L4 in *Streptococcus pneumoniae* have been reported to confer simultaneous resistance to chloramphenicol, oxazolidinones, and macrolides (Wolter et al., 2005). Inactivation of chloramphenicol by *O*-phosphorylation has only been observed in the chloramphenicol producer *Streptomyces venezuelae* and is believed to contribute to self-defense of the host (Mosher et al., 1995).

The gene *estDL136* from a soil metagenome library was found to specify a hydrolase that, when cloned in *E. coli*, inactivated both chloramphenicol and florfenicol (Tao et al., 2012). A chloramphenicol hydrolase gene from the chloramphenicol producer *S. venezuelae* has also been identified and is considered to protect the organism from its antibiotic product (Mosher et al., 1995).

The gene *cfr*, originally detected on a plasmid in a bovine *Staphylococcus sciuri* isolate (Schwarz et al., 2000), codes for an rRNA methylase that targets the adenine residue at position 2503 in 23S rRNA (Long et al., 2006). This adenine residue is located in the overlapping ribosomal binding site of phenicols, lincosamides, oxazolidinones, pleuromutilins, and streptogramin A ($PhLOPS_A$) antibiotics. The additional Cfr-mediated methylation at this site confers resistance to the $PhLOPS_A$ drugs by interfering with their binding to the ribosome. The *cfr* gene is commonly located on plasmids that play a role in its distribution across species and genus boundaries (Shen et al., 2013). The *cfr*(B) gene has been described in *Clostridium difficile* and *Enterococcus faecium* (Deshpande et al., 2015; Hansen and Vester, 2015; Marín et al., 2015).

Distribution of Resistance Genes

Many agricultural and environmental studies are moving away from culturing bacteria and determining what ARGs they carry, using polymerase chain reaction (PCR) or/and quantitative PCR (qPCR) molecular techniques. These molecular assays are often used for direct detection ARGs in food (Guarddon et al., 2011), animal feeding facilities (Stine et al., 2007), and agricultural soils amended with manure (Peng et al., 2015) and used as indicators for water quality changes (Harnisz et al., 2015), among other uses. In each of these studies, known genes were used without necessarily determining their probable distribution in the particular sample source. Other groups are taking a different approach by looking at the microbial resistome of different environments (Forsberg et al., 2012; Perry et al., 2014). The knowledge of which types of bacteria carry which resistance gene(s) is important for future studies so that these resistance genes can be selected to ensure that the whole spectrum of bacteria in the particular ecosystem are included in future surveillance studies (Pei et al., 2006; Smith et al., 2004).

Distribution of Tetracycline-Resistance Genes

Currently there are 50 Gram-positive, cell-wall-free, *Mycobacterium*, *Nocardia*, or *Streptomyces* genera and 76 Gram-negative genera included in Table 1. These include strains that carry one or more of the different *tet*/*otr* resistance genes. Both aerobic and anaerobic genera are included. New *tet* genes and new mosaic gene combinations have been identified (Abril et al., 2010; Palmieri et al., 2012; Warburton et al., 2013). Among the efflux genes, 14 are found only in Gram-negative genera, 12 different genes only in Gram-positive genera, and 4—*tet*(K), *tet*(L), *tet*(39), *tet*(42)—in both Gram-positive and Gram-negative genera (Table 1). The *tet*(K), *tet*(L), and *tet*(39) efflux genes have a G+C content of 28 to 40%, suggestive of a Gram-positive ancestral source, while *tet*(42) has a G+C content of 68% and its ancestral source is unclear.

The *tet*(B) gene has the widest host range among the Gram-negative genera ($n = 33$) and is specific for Gram-negative aerobic and facultative anaerobic bacteria. Thus, if there is interest in determining the Gram-negative tetracycline gene efflux levels in an environmental sample, a *tet*(B) gene should be included in any assay. In contrast, if both Gram-positive and Gram-negative bacteria carrying tetracycline efflux genes are of interest, then the *tet*(L) gene should be used because it is found in 22 different Gram-positive and 19 Gram-negative genera. However, it is less common in aerobic and facultative anaerobic Gram-negative genera but found in strict anaerobic genera, unlike the *tet*(B) gene (Table 1). If other aerobic and facultative anaerobic Gram-negative efflux genes are of interest, the *tet*(A) gene is the next most common, being present in 23 Gram-negative genera, followed by *tet*(D) in 22 genera. Among the 12 Gram-positive *tet*/*otr* genes, three are found only in *Streptomyces*, one in both *Mycobacterium* and *Streptomyces*, one in *Mycobacterium*, one in *Clostridium*, and the remaining six genes are found in two or three Gram-positive genera. However, limited surveillance studies have been done with these 12 efflux genes.

A number of different *tet* efflux genes have been identified in *Aeromonas* ($n = 7$), *Escherichia* ($n = 8$), *Pseudomonas*, and *Vibrio* ($n = 8$ genes each) (Table 1). One study found a correlation between the plasmid incompatibility group and a particular *tet* gene that the plasmid carried (Jones et al., 1992). There are also some *tet* efflux genes, such as *tet*(E), that are associated with genera found in the environment rather than in people or animals. The *tet*(C) gene has been found in the obligate intracellular bacteria *Chlamydia suis* chromosome (Dugan et al., 2004). This marks the first description of any acquired antibiotic resistance gene identified in obligate intracellular bacteria. Associated with the *tet*(C) gene was an IS*CS605* element similar to those found in *Helicobacter*. In addition, a 10.1-kilobase (kb) fragment shared 99% identity between the *C. suis* genetic island and the *Aeromonas salmonicida* plasmid pRAS3.2. One can only speculate how the *tet*(C) gene came to be linked to a genetic island that has components from two distantly unrelated genera that

inhabit distinctive ecological niches. However, these data suggest that even obligate intracellular bacteria like *Chlamydia* spp. are able to exchange and acquire tetracycline-resistance genes. The *tet*(H) gene has been found in 12 genera, but these genera are found in the environment, food-producing animals, humans, and uncultured sludge (Fig. 1) and from bacteria isolated from around the world (Miranda et al., 2003).

Twelve ribosomal protection genes have been identified, and most are of Gram-positive origin. Eight of these genes have been identified in Gram-positive and Gram-negative genera (Table 1). The *tet*(M) and *tet*(Q) genes are generally associated with conjugative chromosomal elements, which code for their own transfer (Roberts, 2005a). Conjugative transposons appear to have less host specificity than do plasmids, which may explain why the *tet*(M) gene is found naturally in 75 different genera (including 38 Gram-positive and 37 Gram-negative genera). The *tet*(O) gene has been found in 35 genera (18 Gram-positive and 17 Gram-negative genera), while the *tet*(W) gene has been identified in 32 different genera (10 Gram-positive and 22 Gram-negative genera). The *tet*(Q) gene has been found in 19 genera (8 Gram-positive and 11 Gram-negative genera), while the *tet*(S) and *tet*(T) genes have been detected in nine genera each (Table 1). The remaining *tet* genes are found in one to three Gram-positive genera (Table 1).

The *tet*(X) gene is found in anaerobic *Bacteroides* spp., although it is unlikely to have much clinical relevance in this host. The *tet*(X) is now in 11 different Gram-negative genera, most of which are associated with the environment, and might be a very interesting gene to use when screening various environmental environments. This mechanism of resistance was considered an oddity until the identification of a second gene, *tet*(37), with the same mechanism of action although genetically unrelated (Diaz-Torres et al., 2003). Unfortunately, the host(s) of the *tet*(37) gene is not known, while *tet*(34), another gene that enzymatically changes the tetracyclines, has been identified in four different Gram-negative bacteria primarily found in the environment.

The *tet*(U) gene has now been identified in three Gram-positive genera. More work with this gene and its associated protein is needed to understand if the protein does code for tetracycline resistance and what that mechanism of resistance is.

Distribution of Phenicol-Resistance Genes

A wide distribution of the classical *cat* genes has been identified for 7 of the 22 groups (Table 2). The Tn*9*-borne *catA1* (*catI*) gene has been found in seven genera of Gram-negative bacteria. Besides chromosomal locations, the *catA1* gene is often detected on large plasmids that carry additional resistance genes. The plasmid-borne gene *catA2* (*catII*) is frequently associated with Cm^r *Haemophilus* spp. (Roberts et al., 1982) but has also been found on plasmids from *Photobacterium damselae* and *Agrobacterium tumefaciens*. The gene *catA3* (*catIII*) from *Shigella flexneri* has also been detected in bacteria different from Enterobacteriaceae. This gene represents part of plasmid-borne multi-resistance gene clusters in *Mannheimia* spp. (Kehrenberg and Schwarz, 2001) and uncultured eubacteria. The staphylococcal *cat*-carrying plasmids pC221, pC223/pSCS7, and pC194 are small plasmids of <5 kb in size that only mediate chloramphenicol resistance (Lyon and Skurray, 1987). These small plasmids have also been isolated

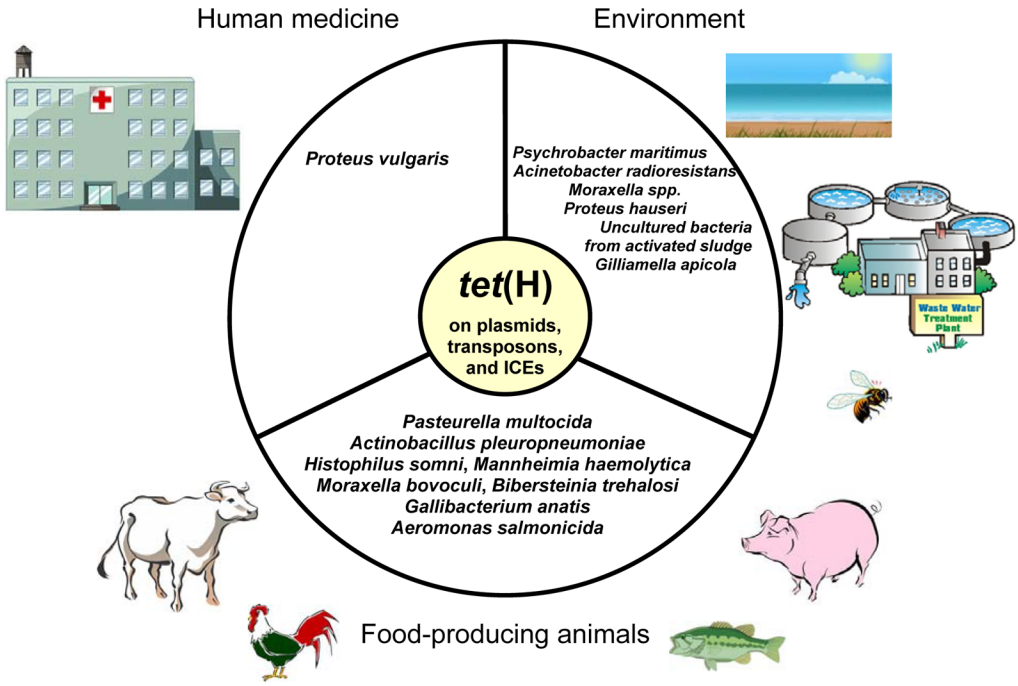

Fig. 1. Distribution of the tetracycline resistance gene *tet*(H), which has been identified as part of transposons (e.g., Tn*5706*) and integrative and conjugative elements (ICEs) (e.g., ICE*Pmu1* and ICE*Mh1*) but also on various plasmids that may also carry additional resistance genes. The *tet*(H) gene has been identified in *Proteus vulgaris* ATCC49132 from a urinary tract infection in a human, the porcine respiratory tract pathogens *Pasteurella multocida* and *Actinobacillus pleuropneumoniae* from Europe and North America, *Moraxella boviculi* from keratoconjunctivitis in cattle of unknown geographical origin, the bovine respiratory tract pathogens *Pasteurella multocida*, *Mannheimia haemolytica*, *Bibersteinia trehalosi*, and *Histophilus somni*, the poultry pathogen *Gallibacterium anatis* from Mexico, and the fish pathogen *Aeromonas salmonicida* from Canada. Moreover, it has been found in *Gilliamella apicola*, a gut symbiont of honey bees and bumble bees in the United States, in uncultured bacteria from activated sludge in Korea, in *Acinetobacter radioresistans* and *Moraxella* spp. from water samples in salmon farms in Chile, in *Psychrobacter maritimus* from coastal sea ice and sediments of the Sea of Japan, and in the bioelectricity-generating and dye-decolorizing bacterium *Proteus hauseri*.

in *Bacillus* spp. (Lyon and Skurray, 1987). Naturally occurring co-integrates between pC221 and pS194 (a small staphylococcal streptomycin-resistance plasmid) have also been detected. Plasmids similar to pC221 can also recombine with larger plasmids to form new resistance plasmids that have a broader host range, extended transfer abilities, and carry additional resistance genes like the conjugative plasmid pIP501, which has a pC221-like *cat* gene, the macrolide resistance gene *erm*(B)gene and the Tc^r gene *tet*(M) (Macrina and Archer, 1993). The *cat* genes of the pC221 group have been detected primarily in *Staphylococcus* spp., *Streptococcus agalactiae*, *Enterococcus faecalis*, and *Bacillus subtilis*, while most of the *cat* genes of the pC223 group have been found on plasmids from *Staphylococcus* spp., *Listeria monocytogenes*, *Lactococcus lactis*, and *Enterococcus faecium*. The pC194-like *cat* genes have been identified mainly in *Staphylococcus aureus*, *Enterococcus faecium*, *Lactobacillus reuteri*, and *Streptococcus suis*.

The *catB1*gene from *Agrobacterium tumefaciens*, the *catB7* gene from *Pseudomonas aeruginosa*, and the *catB9* gene from *Vibrio cholerae* have been found exclusively on the chromosome. In contrast, the Tn*2424*-borne *catB2* gene has been detected on plasmids from Gram-negative enteric genera and in the chromosome of *Shewanella oneidensis*, an environmental bacterium. The closely related *catB3* to *catB8* genes are usually located on plasmids and are widespread among various Gram-negative genera (Table 2).

The first two groups of specific exporter genes are most widespread among Gram-negative bacteria (Table 2). The cassette-borne *cmlA* group is frequently found on multi-resistance integrons or associated with transposons located on conjugative and/or nonconjugative plasmids in *Pseudomonas aeruginosa* and various enteric genera. The genes *fexA* and *fexB* are most frequently associated with staphylococcal and enterococcal plasmids, respectively. The *cfr* gene has been identified mainly on plasmids but also in the chromosomal DNA of *Staphylococcus* spp., *Bacillus* spp., *Enterococcus* spp., *Macrococcus caseolyticus*, *Jeotgalicoccus pinnepedialis*, *Streptococcus suis*, *Proteus vulgaris*, and *E. coli* (Roberts and Schwarz, 2009; Shen et al., 2013; Wang et al., 2013). Despite their diverse origins, all these Cfr proteins were either indistinguishable from the original Cfr protein from *Staphylococcus sciuri* or differed by at maximum two amino acids (= 99.4% identity). The *cfr*(B) gene has been identified in two Gram-positive genera (Table 2). The *cfr*(B) gene from *Clostridium difficile* (Marín et al., 2015) revealed only 75.1% identity to the original Cfr protein. Most recently, the *cfr*(B) gene was cloned from *Clostridium difficile* and shown to confer multiple antimicrobial resistance by the same mechanism as the original *cfr* gene (Hansen and Vester, 2015). The *cfr*(B) gene has also been found in *Enterococcus faecium* (Deshpande et al., 2015).

It should be noted that independent acquisition of mobile elements carrying *cat*, *cmlA*, *floR*, or *cfr* genes can lead to the simultaneous occurrence of more than one type of phenicol resistance gene in the same bacterium. Thus, multi-resistant isolates of *Salmonella enterica* servar Typhimurium DT104 var. Copenhagen carried a *catA1* gene in addition to a *floR* gene (Frech et al., 2003), while *catA2* and *catA3* genes were detected on the same plasmids in *Klebsiella pneumoniae* (Soge et al., 2006) and *catA3* together with *floR* on a plasmid from *Bibersteinia trehalosi* (Kehrenberg et al., 2006). Plasmids carrying *fexA* and *cfr* or *fexB* and *cfr* have also been described (Roberts and Schwarz, 2009; Shen et al., 2013).

Conclusion

Bacterial resistance to tetracycline and/or phenicols due to the acquisition of new genes and/or the mutation of existing genes has increased during the last 30 yr, especially as environmental sources have been examined. The environment may have many new genes that are not found in bacteria traditionally associated with animals or humans, and it is unclear whether these "new genes" will impact the treatment of animal and human infections in the future. Resistance levels vary by geography and by species, as well as agricultural practices. However, many pathogenic and opportunistic bacteria found in agricultural settings, aquaculture, the environment, and humans are resistant to one or both of these classes of antibiotics. It is clear that ARB and ARGs are spread between animals, the environment, and humans and from one geographic location to another around the world. The environment is clearly an important reservoir for these resistance genes, and more work is needed to better define what influences promote transfer between the environmental microbiome and the animal and human microbiomes.

Acquired genes are often associated with mobile elements that provide flexibility to the host bacteria and help in the spread and distribution of these genes across diverse bacterial populations and ecosystems. Multiple ARGs can be clustered on individual mobile elements, which allows multi-resistance to be transferred by a single genetic element. This can quickly increase the multiple-drug-resistant bacterial populations. Unless the overall use of antibiotics changes, this trend is likely to continue, reducing the efficacy of current therapies as outlined in reports from the Centers for Disease Control and Prevention (2013) and the World Health Organization (2014).

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